PowerTree

Feature Overview — v0.5.0

Electronic circuit power tree analysis · generated 2026-08-07 · fully offline Windows application

PowerTree budgets electrical power from source to load with min/typ/max corner analysis, live margin checking, and report-grade deliverables. One project file holds any number of power trees, operating states, documentation notes and design-review waivers. Four interaction modes share one engine: desktop GUI, command line, Excel, and an MCP server for AI assistants.

Modelling

• One source per tree (V min/typ/max, current- or power-limit) feeding converters, loads and series elements

• Converters: topology, Vout corners, quiescent current, output limits, pass-through output rail — plus datasheet efficiency-vs-load curves interpolated at the solved operating point

• Loads: current- or power-type, typ + peak values, allowed input-voltage window for margin analysis

• Series elements: resistor / ferrite bead / inductor / fuse / cable / connector / switch subtypes with DCR, informational inductance, text rating, optional Vin window and current/dissipation ratings

• Blocks group elements into devices (one IC with its Icc/Iq/bank loads; a regulator with its own controller Iq) with aggregate power display

• Device template library: Zynq-7000 SoC, DDR3, GigE/USB PHYs, clock gen, QSPI, SD, buck/LDO regulator blocks — plus user-defined templates from JSON (no code needed)

• Global net registry: signal names are project-wide; voltage conflicts and double-driven rails are flagged

• Full metadata everywhere: signal name, ref des, part number, pins, datasheet link, notes, linked documentation

Solver & analysis

• Automatic bottom-up recalculation on every edit, three corners (min/typ/max), damped fixed-point for series-R × power-load coupling, bounded math that warns instead of crashing

• Margin analysis: limit usage, voltage-window breaches, series ratings, buck/boost sanity, collapsed rails, net conflicts — flagged but never blocking

• Derating policy (default 80% of any limit) as a project-level design rule

• Rail budgets: remaining headroom per limited rail expressed as extra load available

• Load-growth capacity: the uniform load-scaling the design tolerates before its first violation, with the bottleneck named

• End-to-end efficiency, total conversion/distribution loss, top consumers with % of source budget

• Operating states: named modes with per-element overrides, live state selector, per-state validation, materialize-state-as-tree

• Finding waivers with engineering justification — waived items stop counting but remain in every report as an audit trail

Visualization

• Flowchart canvas: color-coded cards with live power text, bus-routed 90° edges with junction dots on branched rails, block outlines, legend, warning badges

• Layouts: top-down, left-right, or custom drag-to-place (positions saved); collapse/expand any branch

• Heat map: cards tinted cold-blue to hot-red by power draw

• Print style: white ink-friendly palette for canvas and exports

• Display detail cascade (app → tree → element): minimal / standard / exhaustive card verbosity

• List view with electrical columns, % of tree, and status; search (Ctrl+F) across name/refdes/signal/part

Documentation

• Hierarchical markdown notes vault with embedded images, full-text search, element linking

• Per-element documentation editing straight from the Properties panel (markdown + live preview + image import)

• Notes export to Markdown / HTML / PDF — whole vault or any subtree

Reports & sharing

• PDF report: executive summary (KPIs, health verdict, top consumers, growth capacity), global nets, per-tree flowcharts, corner tables, rail budgets, states, findings incl. waivers, notes appendix

• Single-file HTML report (images embedded) — mail or wiki, zero install

• Excel report: outline collapse/expand, live P=V·I formulas, States / Parts / Warnings sheets, VBA navigation macros (.xlsm via COM, .xlsx+.bas fallback)

• HD PNG flowchart export at configurable scale/style; CSV of the solved table; BOM parts list

• Export bundle: every format into one folder in one click

• Clipboard: copy flowchart as image, copy solved table (Excel-ready)

Workflow & application

• Draft-based element creation (Save/Cancel before anything is added); right-click context menus; duplicate-with-subtree; undo/redo

• Autosave every 3 minutes with crash recovery; recent projects; project properties

• Settings dialog: canvas style, card detail, heat, legend, orientation, PNG scale, PDF content, significant digits

• Self-contained versioned .ptproj file format (JSON, diff-friendly, images embedded)

• Onboarding: first-run welcome, F1 quick-start, full user guide

Automation & integration

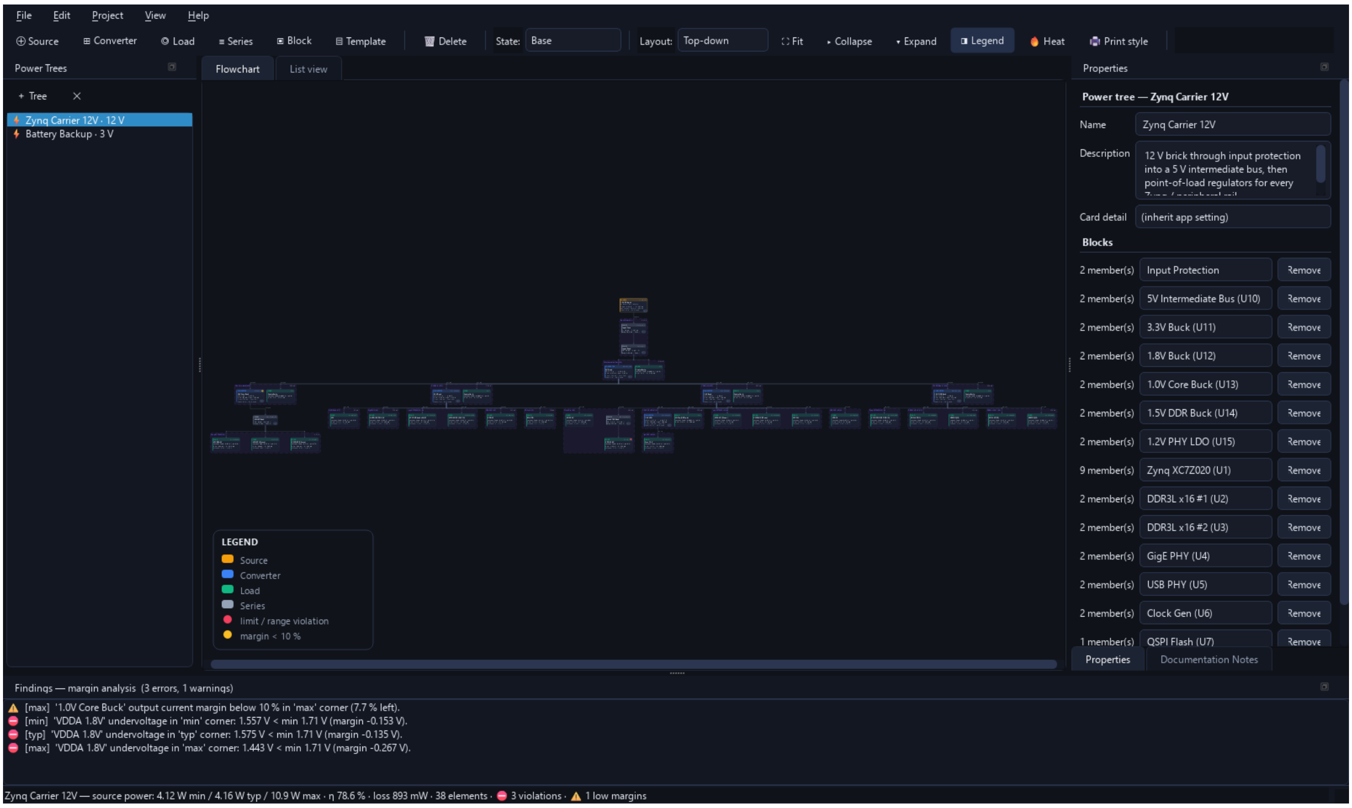
• CLI: info, solve (per state), validate (CI gate, --strict), nets, headroom, growth, bom, search, templates, export, demo, gui — all with --json

• MCP server (16 tools) for AI assistants: open/edit/solve/validate/waive/export projects programmatically

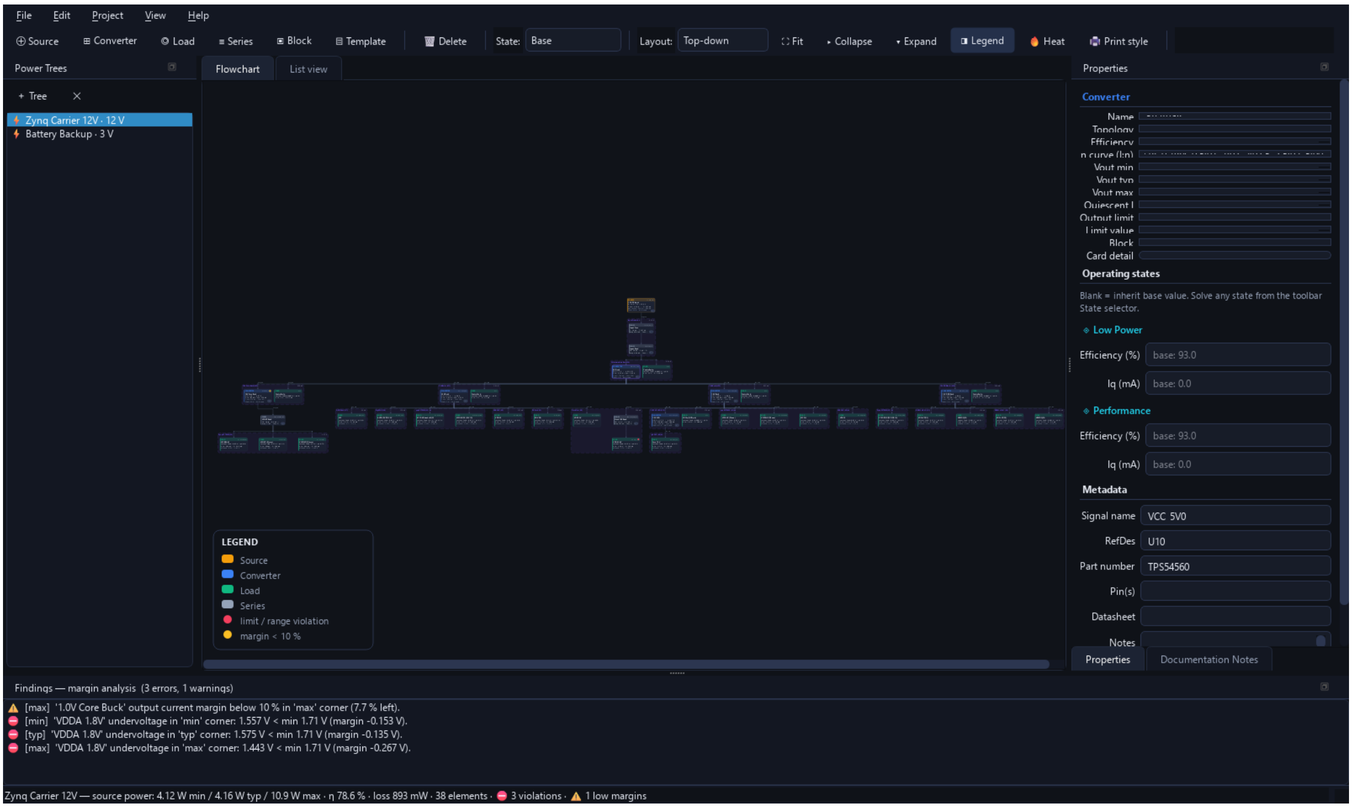
• GitHub Actions CI: full test suite (72 tests + GUI smoke), CLI gate checks, installer artifact build

• Windows installer: PyInstaller app (GUI + CLI exes), per-user install with shortcuts and uninstaller, Inno Setup script, .ptproj file association

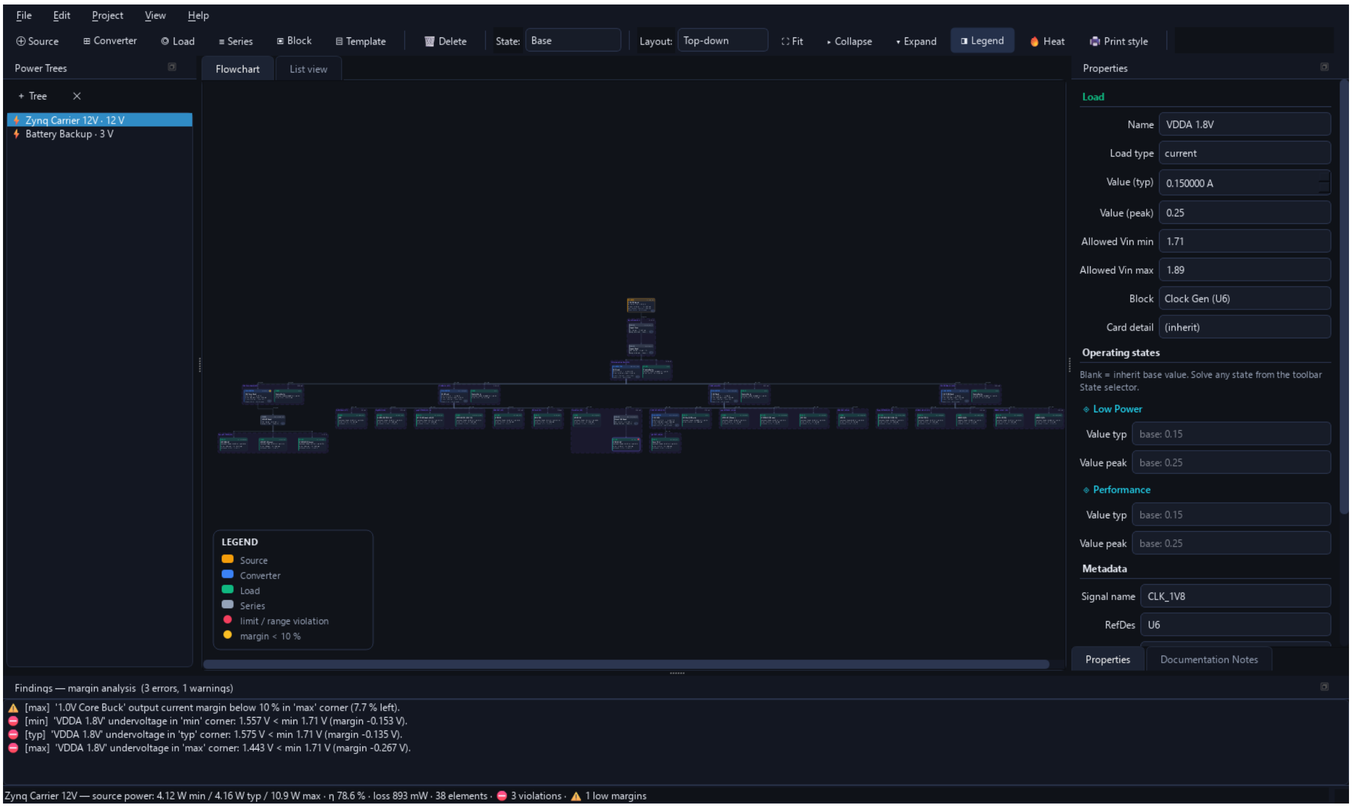
Screenshots



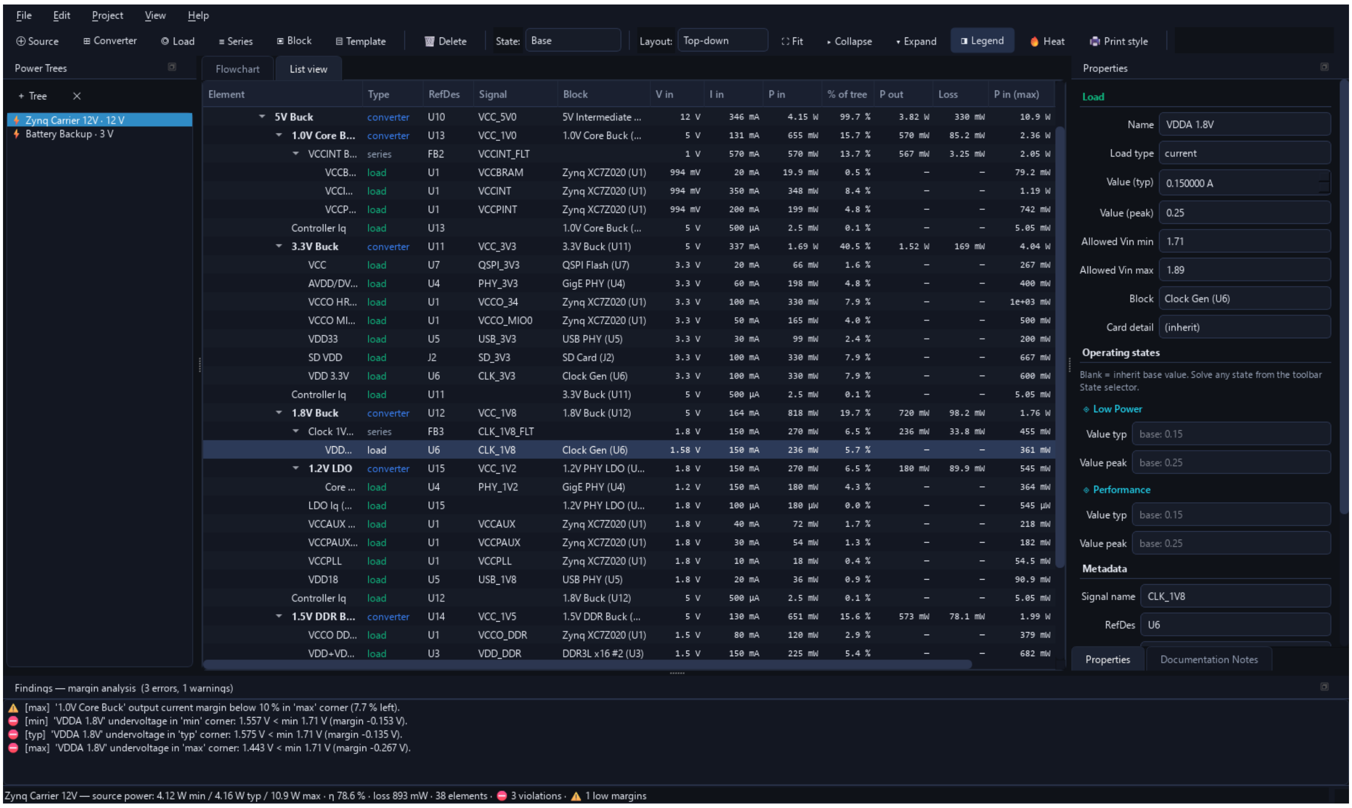
Main window — Zynq carrier demo: flowchart with color-coded cards, block outlines, bus-routed edges, legend, findings dock and live status bar (efficiency, loss, health).



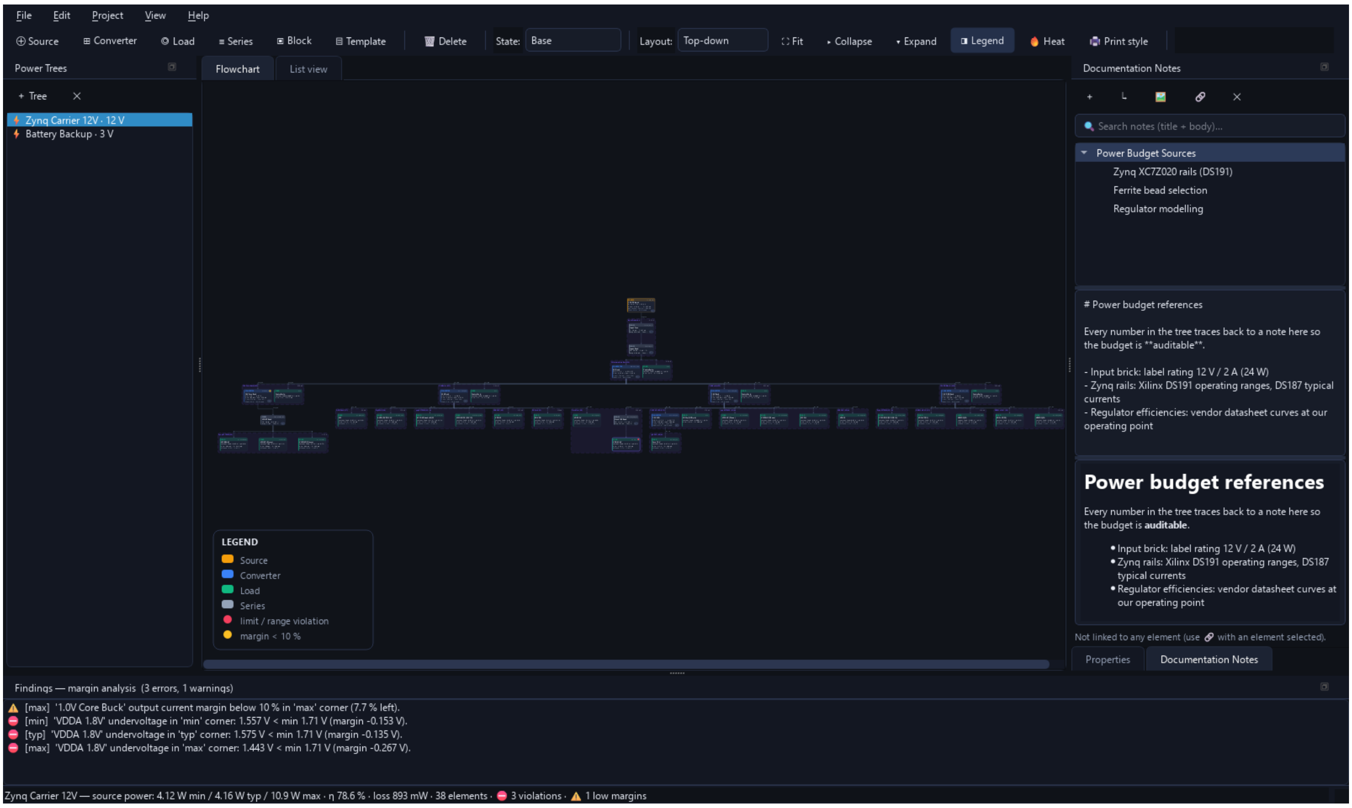
Properties panel for a converter: Vout corners, efficiency + datasheet η-curve, limits, per-state overrides, metadata, block assignment, card-detail override, documentation buttons.



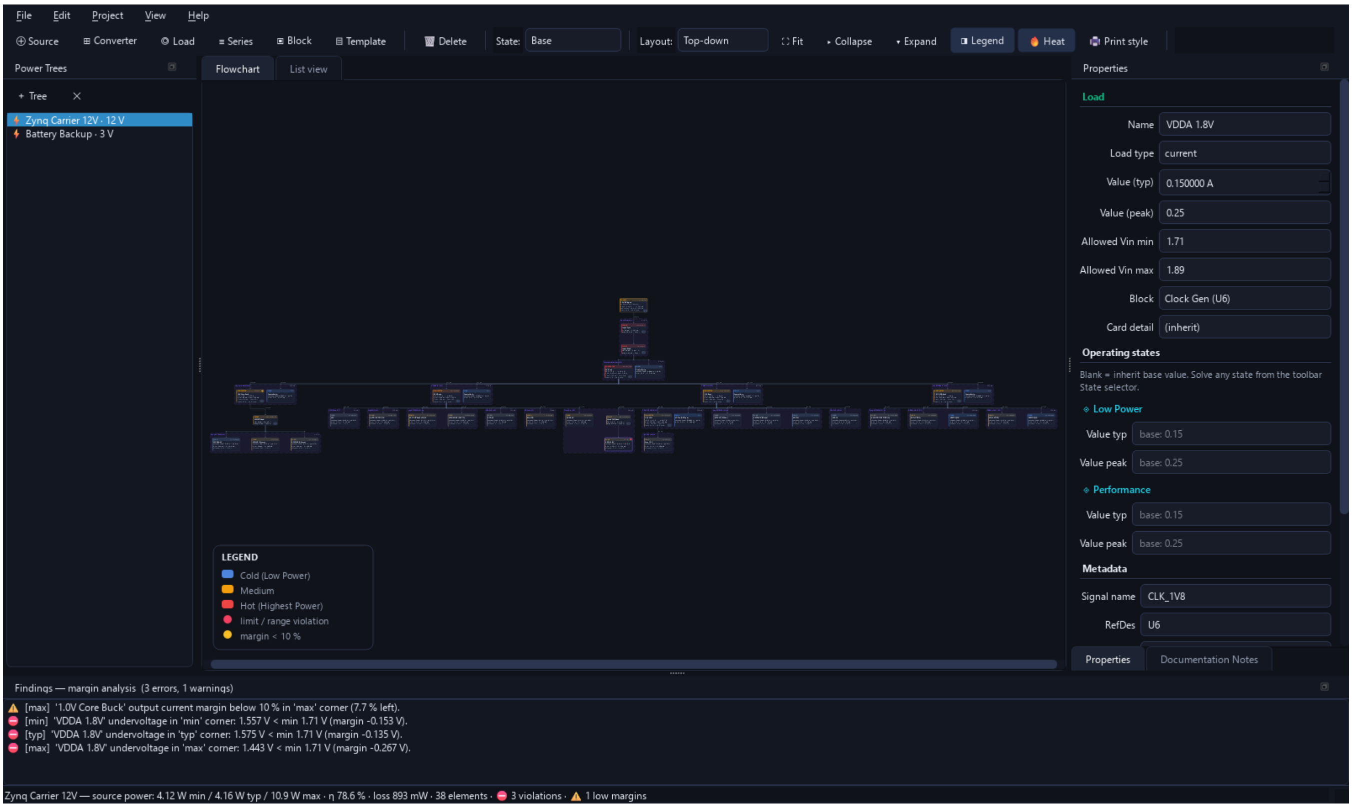
A load with a live margin finding — the wrong ferrite bead drops the clock rail below its window: red badge on the card, entry in the Findings dock, allowed-window fields in Properties.



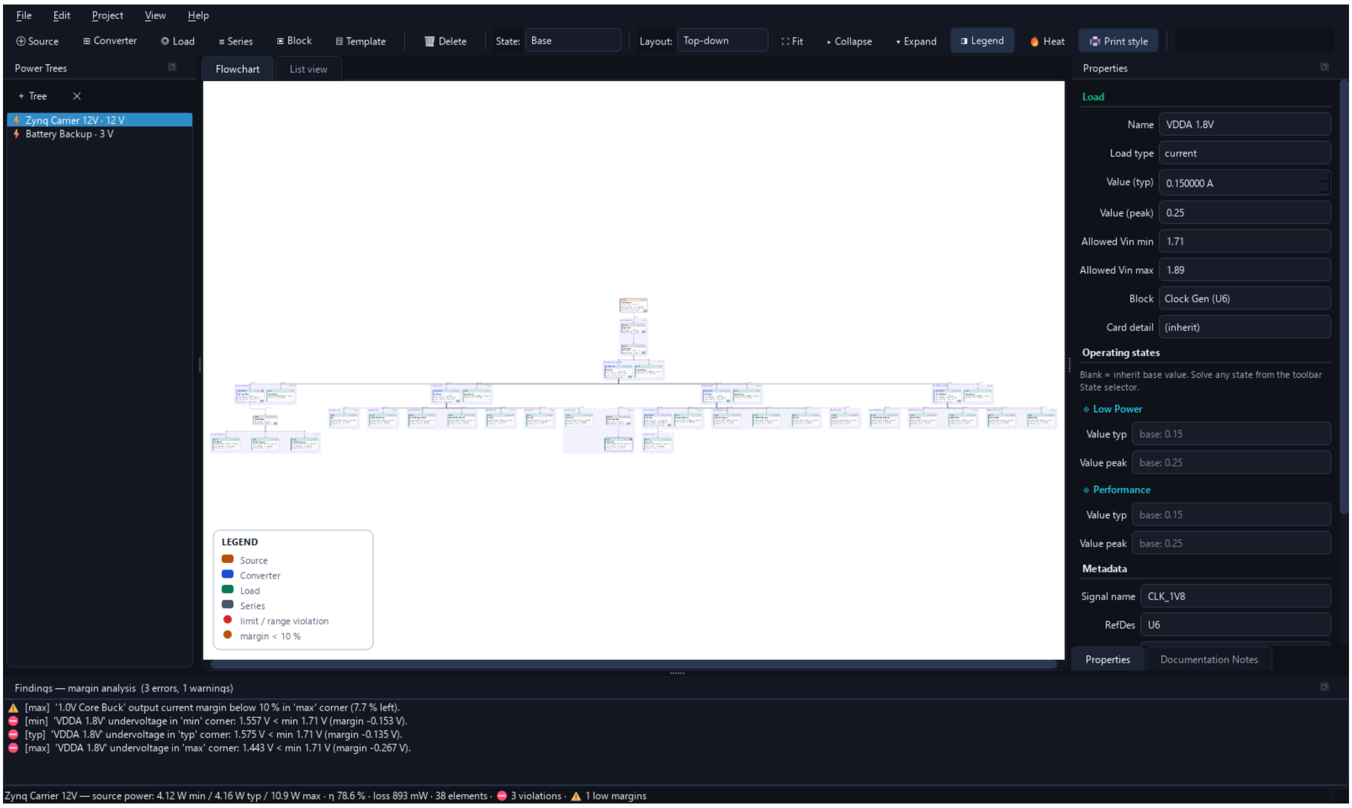
List view: full hierarchy with V/I/P, % of tree, losses, worst-case power and status per element.



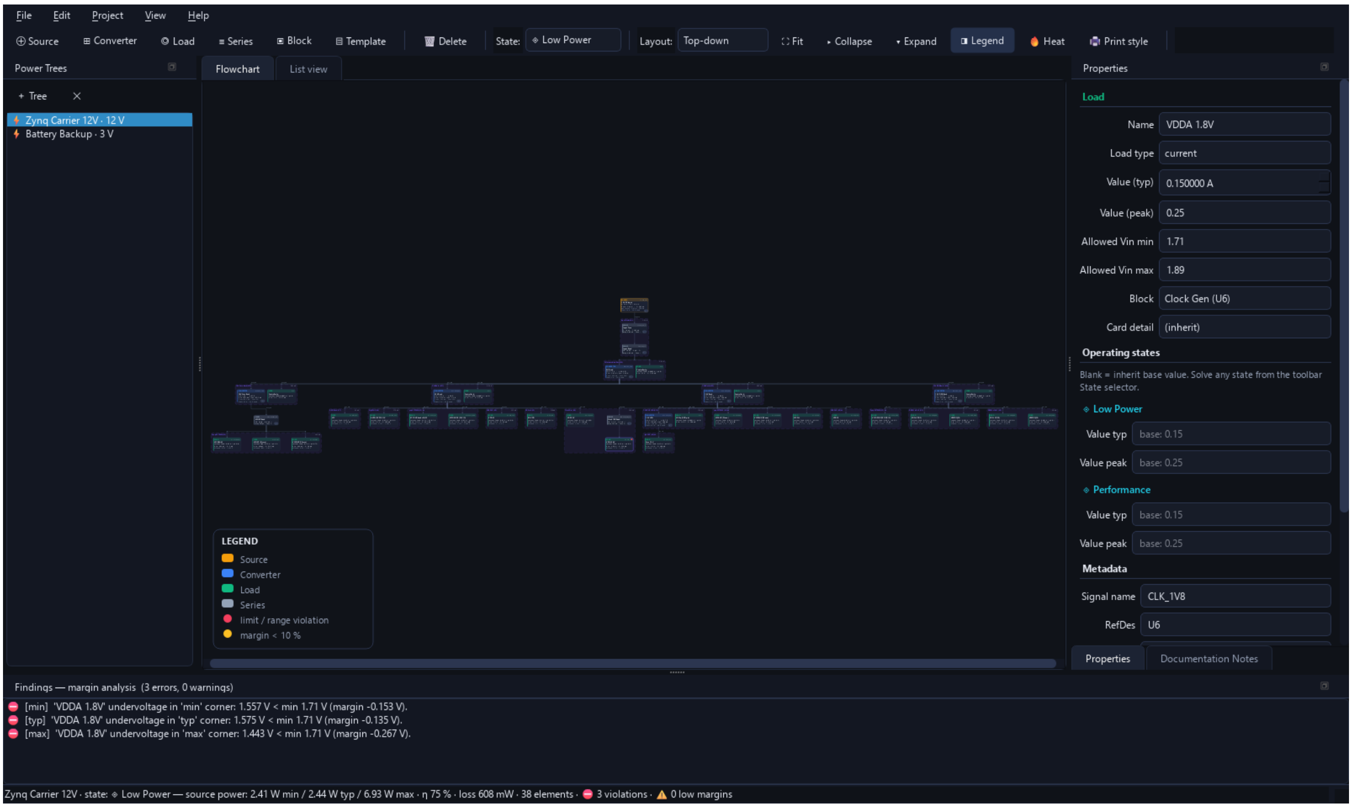
Documentation notes vault: hierarchical markdown notes with live preview, images, element links and full-text search.



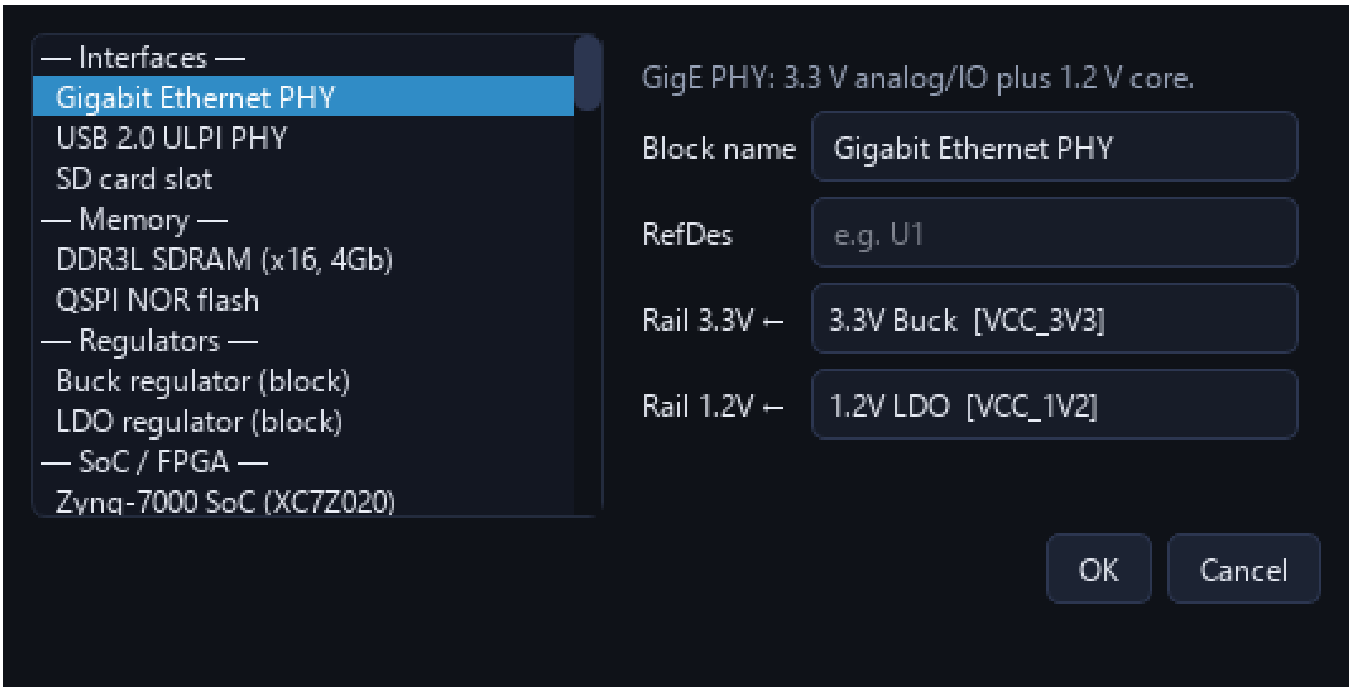
Heat map mode: every card tinted cold-blue → hot-red by its power draw; the legend switches to the heat scale.



Print style: white ink-friendly palette — also used by image/PDF exports while active.



Operating state 'Low Power' selected: the whole tree re-solves with per-element state overrides (source power drops accordingly).



Device template dialog: pick a device (Zynq SoC, DDR3, PHY, regulator block…), map its rails to existing elements, done. User JSON templates appear here too.

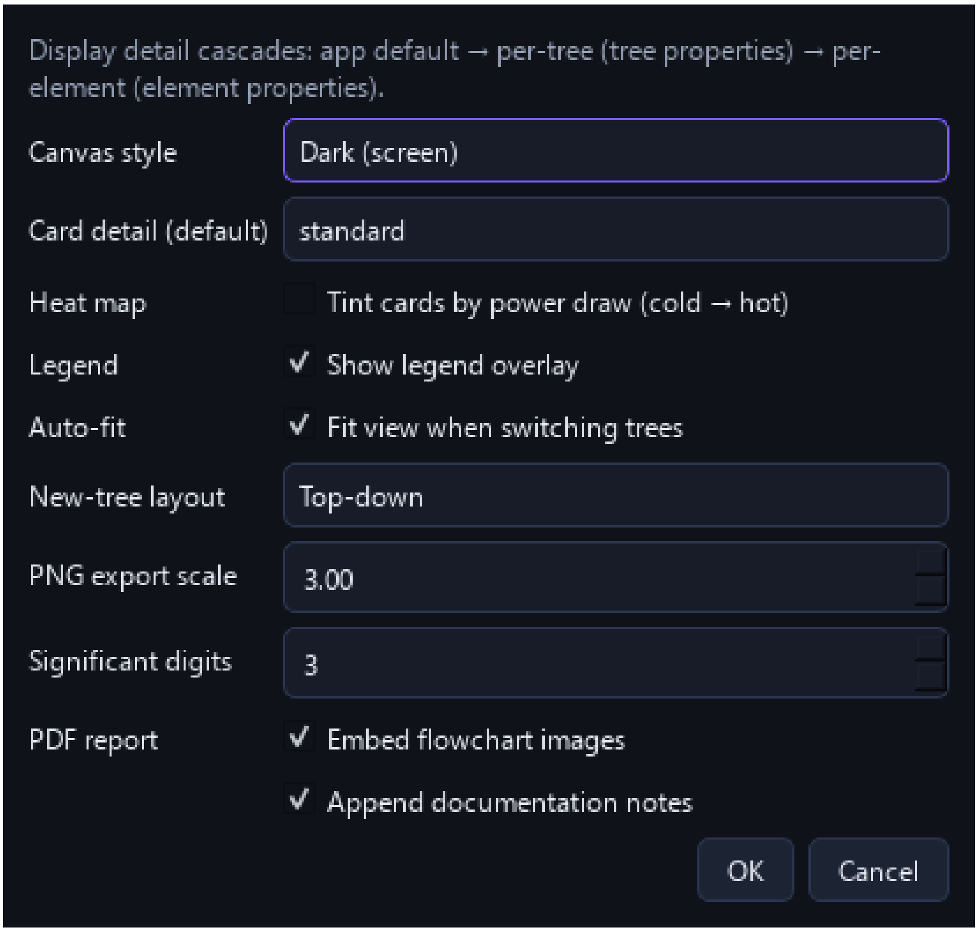

Net (signal) names are global to the project: the same name in any tree is the same electrical node. Sources, converter outputs and series-element outputs define nets; loads consume them.

Net	V typ	Defined by	Loads fed
CLK_1V8_FLT	—	Clock 1V8 Bead (Zynq Carrier 12V)	1
VBAT	3 V	CR2032 Coin Cell (Battery Backup)	0
VBAT_RTC	—	Battery ESR + Diode (Battery Backup)	1
VCCINT_FLT	—	VCCINT Bead (Zynq Carrier 12V)	3
VCC_1V0	1 V	1.0V Core Buck (Zynq Carrier 12V)	0
VCC_1V2	1.2 V	1.2V LDO (Zynq Carrier 12V)	1
VCC_1V5	1.5 V	1.5V DDR Buck (Zynq Carrier 12V)	5
VCC_1V8	1.8 V	1.8V Buck (Zynq Carrier 12V)	5
VCC_3V3	3.3 V	3.3V Buck (Zynq Carrier 12V)	7
VCC_5V0	5 V	5V Buck (Zynq Carrier 12V)	4
VIN_12V	12 V	12V DC Input (Zynq Carrier 12V)	0
VIN_12V_F	—	Input Fuse (Zynq Carrier 12V)	0
VIN_FLT	—	Input Bead (Zynq Carrier 12V)	1

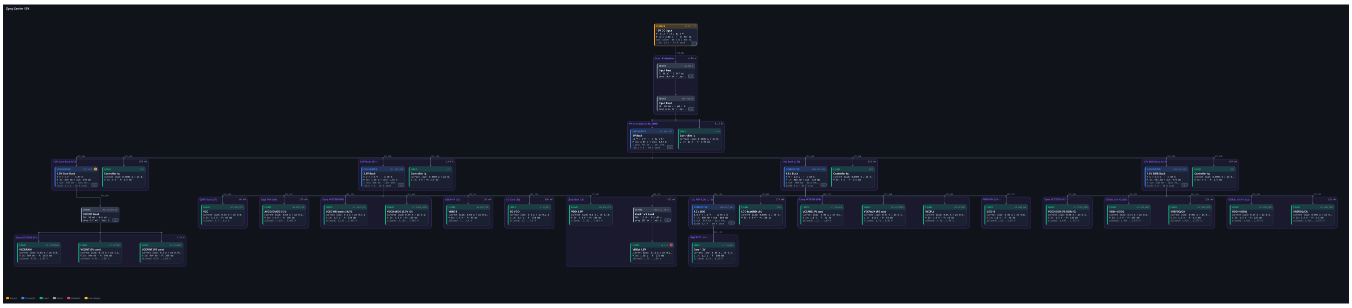
✅ No net conflicts — every net has a single consistent definition.

Close

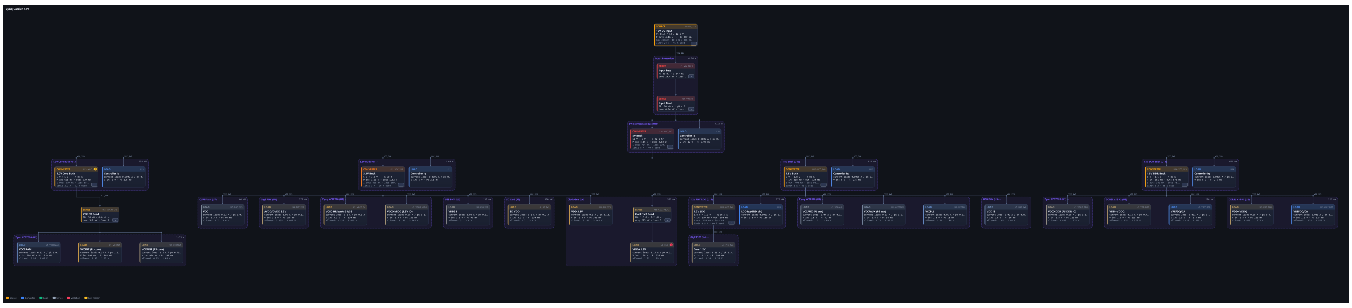
Global net registry: every named rail, its definers, nominal voltage, loads fed — with conflict detection.



Settings: style, card detail cascade, heat, auto-fit, export options, significant digits.



HD flowchart export (dark) — the same renderer drives the canvas, PNG/PDF/HTML exports and the clipboard copy.



HD flowchart export with heat map — instantly shows where the power goes.